The coherence type over a right inverse of a half adjoint equivalence is contractible

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 351-353 and its proof environment opens on line 354, the next line; the pairing is by adjacency and the retained proof range 354-357 is the whole environment.

Editorial note. The retained counter values (chapter 4, section 2, theorem offset 11) make the counted lemma print as Lemma 4.2.12, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:coh-hprop = 4.2.12).

Editorial note. Two labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: lem:coh-hfib (book Lemma 4.2.11, defined at line 340 of this chapter pinned file) and thm:contr-hae (book Theorem 4.2.6, defined at line 265 of the same file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: contractibility of the coherence type of a half adjoint equivalence (author label lem:coh-hprop, book Lemma 4.2.12).

Allowed standard prerequisites (editorial).

- Half adjoint equivalence and right inverses. A half adjoint equivalence on f consists of an inverse with a homotopy, and over each right inverse of f there is a type of coherence data relating the two triangle identities.. They are the hypothesis and the type shown contractible.
- Coherence as a path in a fiber. The coherence type is equivalent to a dependent product over the domain of path types inside the fibers of f.. It reduces the goal to a pointwise statement.
- Contractibility of the fiber and of its path spaces. In a half adjoint equivalence the fiber of f over a point is contractible, and a path space of a contractible type is contractible.. They settle each pointwise goal.

Source reference. Original lemma 4.2.11: sources/hott-equivalences/pinned-original-source.tex, source lines 340–340. Book Lemma 4.2.11, the identification of the coherence type with a dependent product of fiber path types, used at the first line of the retained proof. Navigation pointer only.

Source reference. Original theorem 4.2.6: sources/hott-equivalences/pinned-original-source.tex, source lines 265–265. Book Theorem 4.2.6, contractibility of the fibers of a half adjoint equivalence, used at the second line of the retained proof. Navigation pointer only.

Lemma 4.2.12. *If f is a half adjoint equivalence, then for any $(g, \epsilon) : \mathsf{rinv}(f)$, the type $\mathsf{rcoh}_f(g, \epsilon)$ is contractible.*

Proof. By original lemma 4.2.11 and the fact that dependent function types preserve contractible spaces, it suffices to show that for each $x : A$, the type $(gfx, \epsilon(fx)) =_{\mathsf{fib}_f(fx)} (x, \mathsf{refl}_{fx})$ is contractible. But by original theorem 4.2.6, $\mathsf{fib}_f(fx)$ is contractible, and any path space of a contractible space is itself contractible. □